Python Basics: input() and Control Flow

- Topics Covered:
  - - input() Function
  - - Control Flow in Python
  - - Conditional Statements: if, elif, else
  - - Nested Conditions

input() Function

- Definition: Takes user input during execution.
- Syntax: user_input = input("Enter something: ")
- Example:
  - name = input("Enter your name: ")
  - print("Hello", name)
  - Note: input() returns a string.

Type Conversion with input()

- Convert input to required data type.
- Example:
  - age = int(input("Enter your age: "))
  - print("You will be", age + 1, "next year.")

What is Control Flow?

- Control Flow = Decision-making in programs.
- Python runs code top to bottom.
- Control flow allows:
  - - Making decisions
  - - Repeating actions
  - - Skipping code blocks

if Statement

- Syntax:

```
if condition:
    # code block
```

- Example:

```
age = int(input("Enter your age: "))
if age >= 18:
    print("You are eligible to vote.")
```

if...else Statement

- Syntax:

```
if condition:
    # true block
else:
    # false block
```

- Example:

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

if...elif...else Statement

- Syntax:

```
if condition1:
    # block1
elif condition2:
    # block2
else:
    # default block
```

Nested Conditions

- Definition: if inside another if/else/elif.
- Example:

```
x = int(input("Enter a number: "))
if x > 0:
    if x % 2 == 0:
        print("Positive Even")
    else:
        print("Positive Odd")
else:
    print("Not a positive number")
```

Best Practices

- - Use proper indentation.
- - Avoid deeply nested conditions.
- - Use logical operators when possible.
- - Keep code simple and readable.

Programs

- Write a program to print yes when the number entered by the user is greater than or equal to 25.
- Write a program to find the greatest of four numbers entered by the user.
- Write a program to find whether a given number is even or odd
- Write a program to find out whether a student has passed or failed if it requires a total of 50% and at least 40% in each subject to pass. Assume 4 subjects and take marks as an input from the user.
- Write a program to find whether a given username contains less than 8 characters or not.
- Write a program which finds out whether a given name is present in a list or not.

- Write a program to calculate the grade of a student from his marks from the following scheme:

  90 – 100 => Distinction

  80 – 90 => A+

  70 – 80 => A

  60 – 70 =>B

  50 – 60 => C

  <50 => D